Dynamic Graph Algorithms

CSC-531

Incremental DFS in Undirected Graphs

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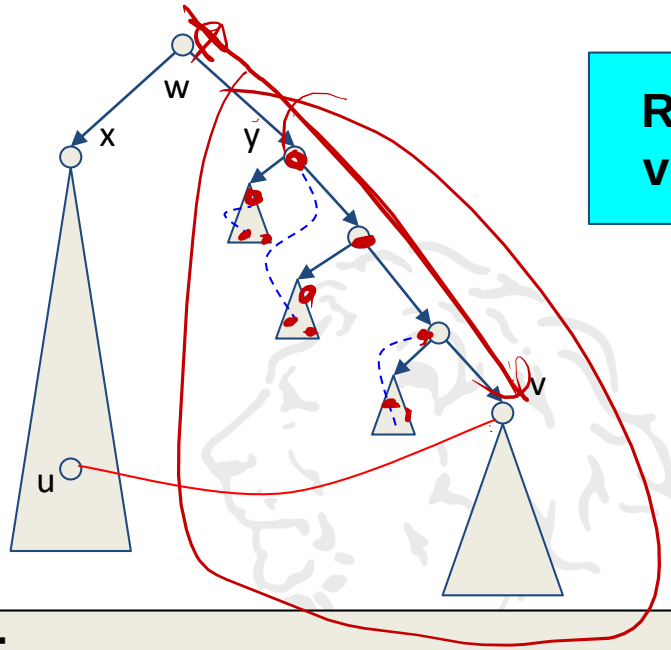
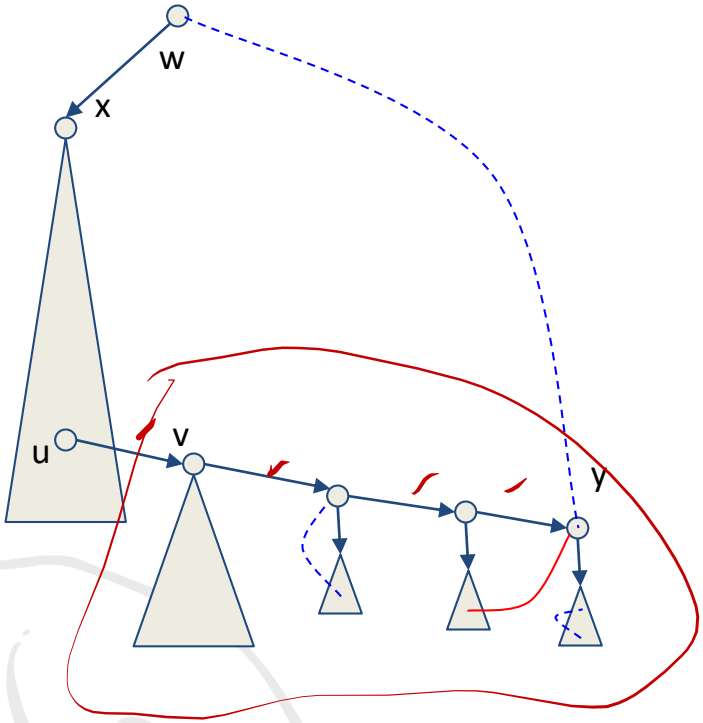
Incremental Depth First Search Trees (Undirected)



Reverse v-y path



Algorithm:

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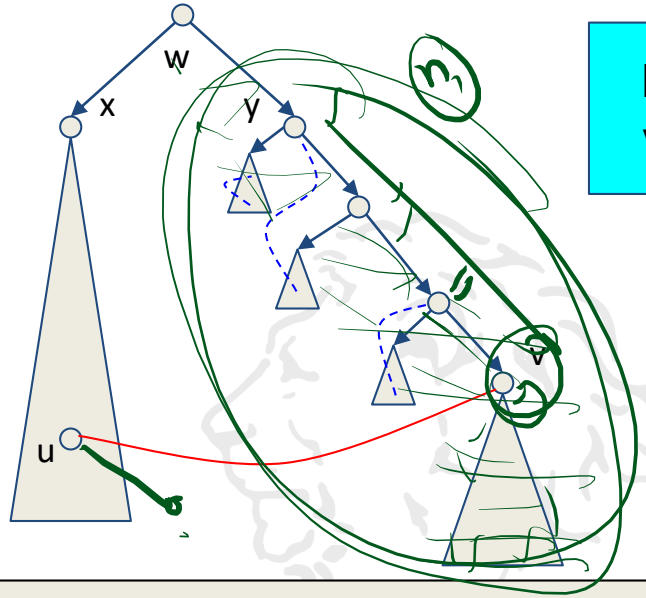
        Add (u,v) to DFS Tree
        Add cross edges of T(v) in InsE
        Update LCA structure → O(T(v))

Analysis?

Collect all Non-tree edges in InsE

Incremental Depth First Search Trees (Undirected)



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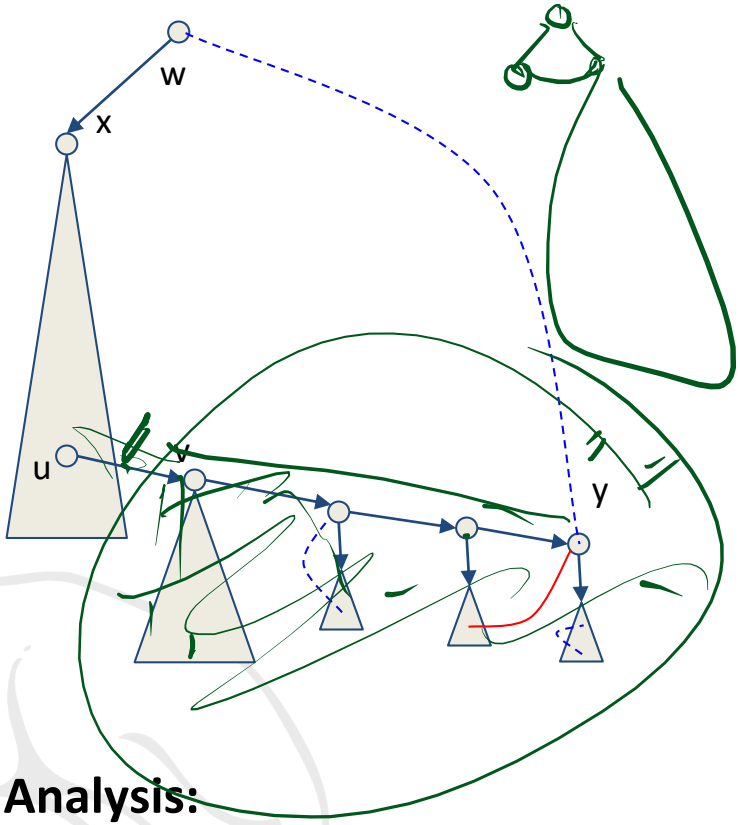

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O(1)

Analysis:

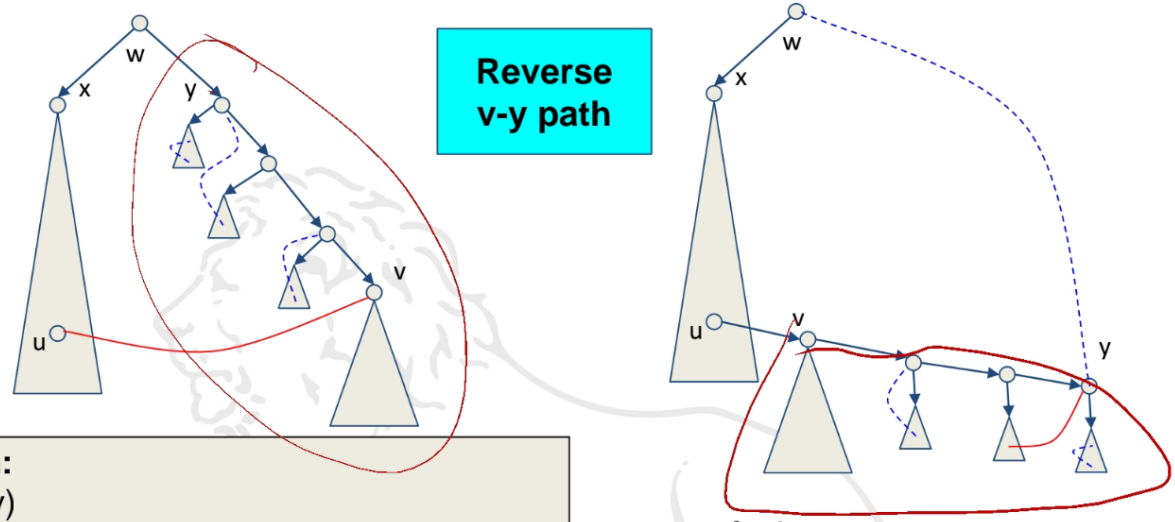
deg(x) whenever x in T(v)

Or deg(x) whenever x falls

WC = O(m+n)

Update LCA structure

Incremental Depth First Search Trees (Undirected)



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Analysis:

deg(x) whenever x in T(v)

Or deg(x) whenever x falls

Total= O(n) ∑ deg(v) = O(mn)

Amortized O(n)

Incremental Depth First Search Trees (Undirected)

Suppose no-cross edges needed to be collected

↳ Cost is $|T(y)|$

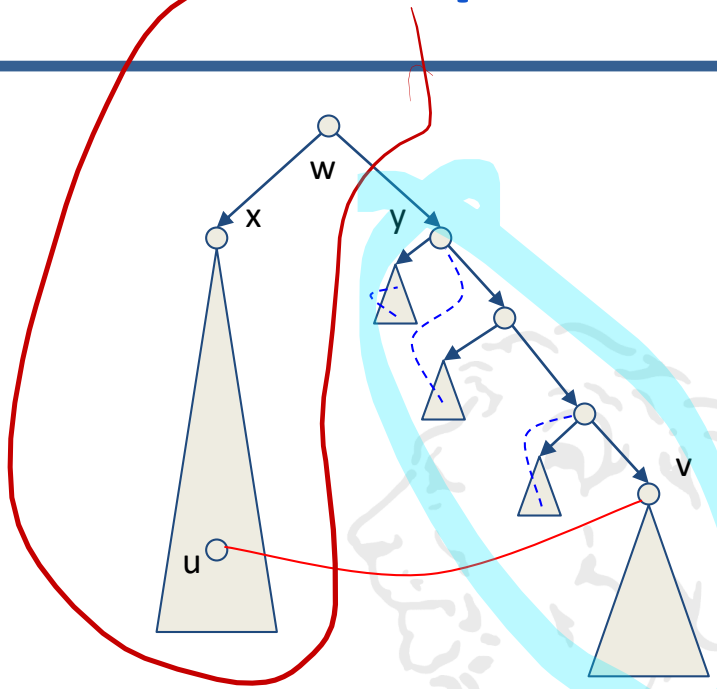
WC $O(n)$

Am. $O(n^2/m)$ → Dense $O(1)$

Total $O(n) \times \sum_{v \in V} 1$

$= O(n^2)$

Incremental Depth First Search Trees (Undirected)



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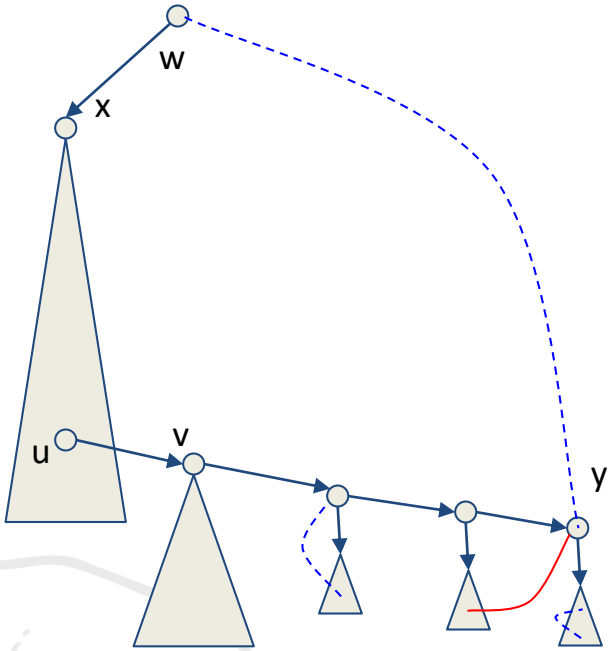

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Correctness?

Every non-tree edge is a back edge

Induction

Inv G \ InsE has no cross edge

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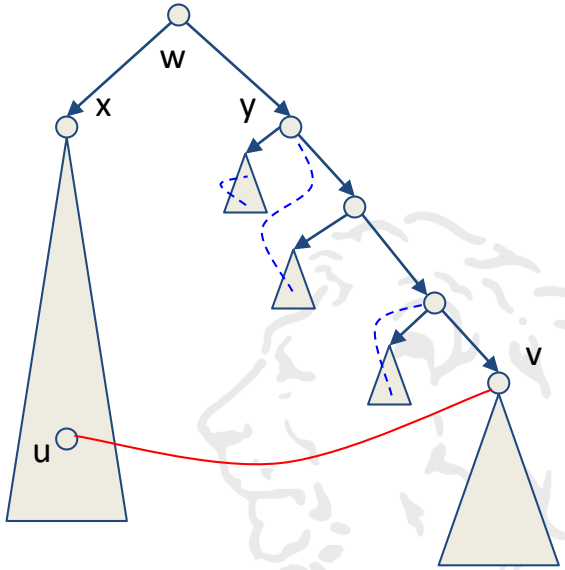

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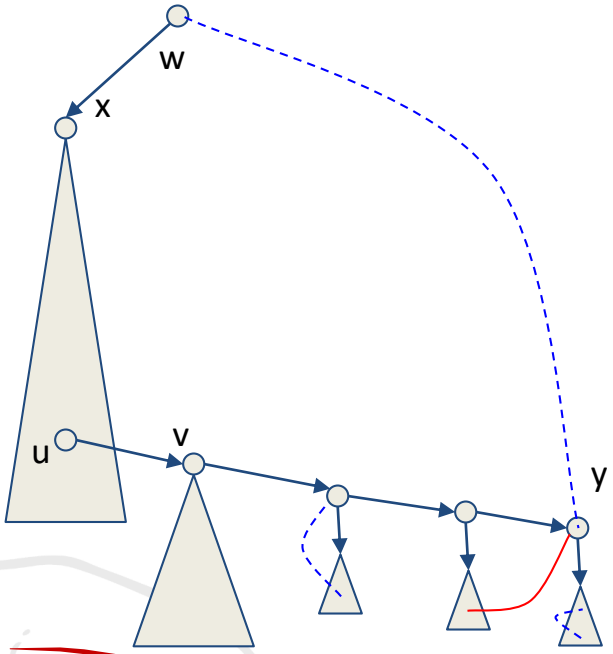
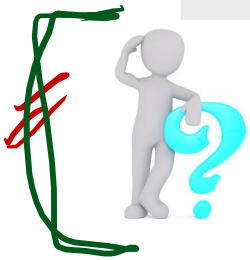


Approach: O(mn)
Collect the Cross Edges
Insert them one by one

Improve to O(n2)



Think?
How does this cross edge look like?
How many such cross edges?

Back Edges converting to Cross Edge

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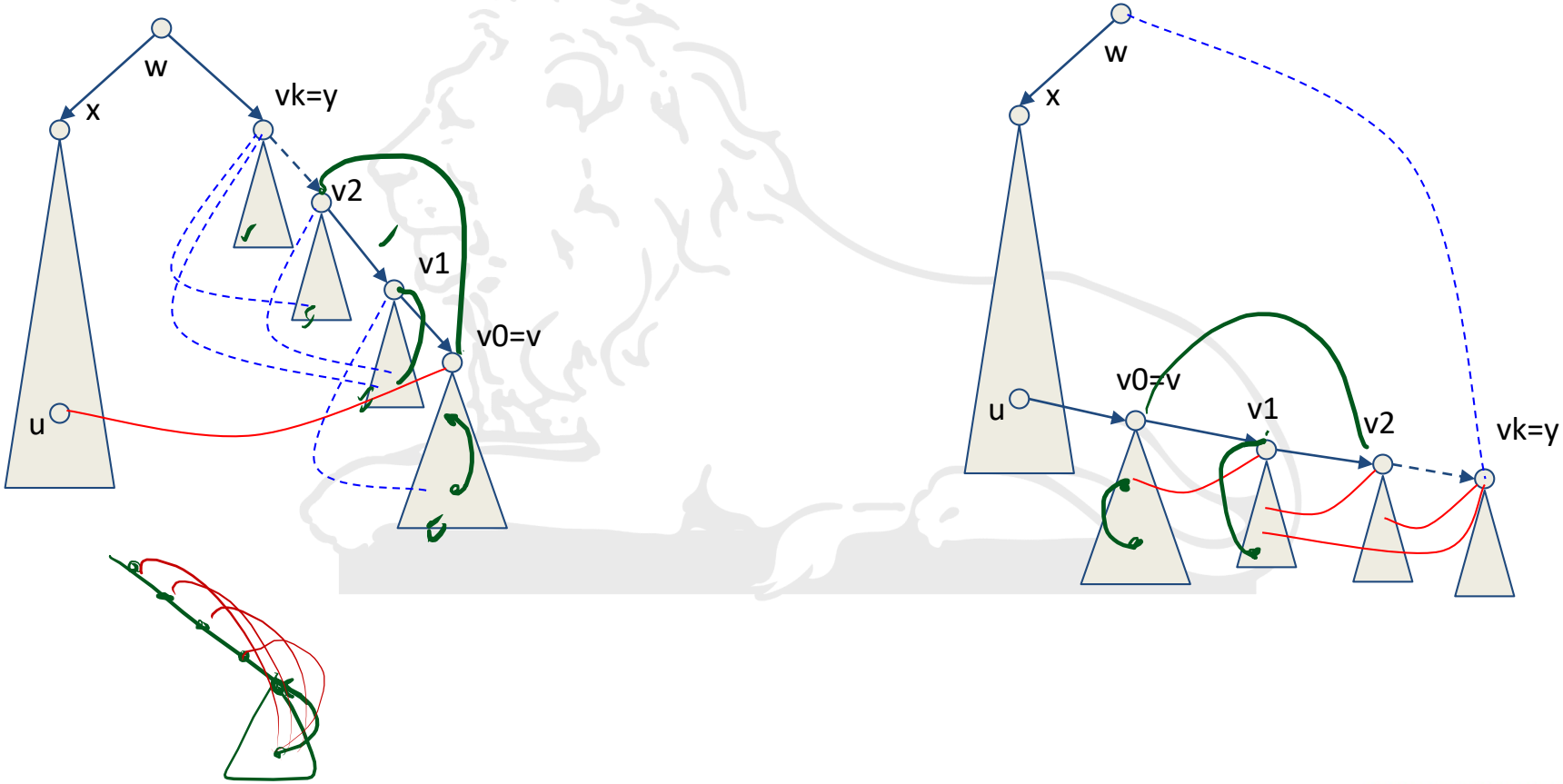

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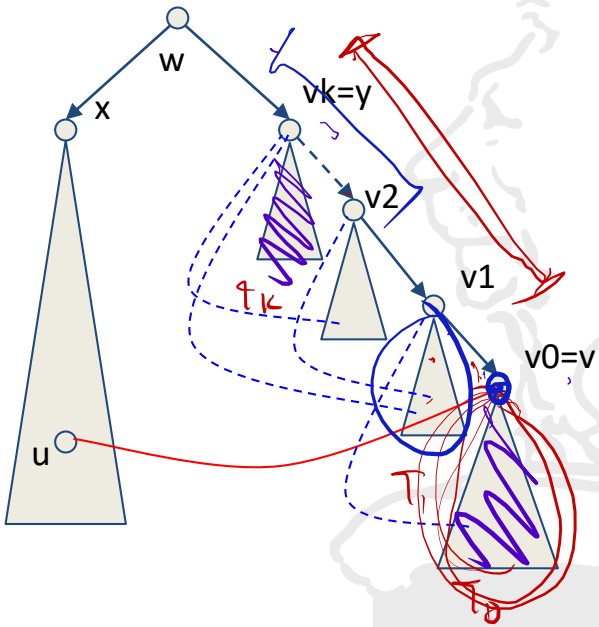


Edges between Ti and vj, j>i

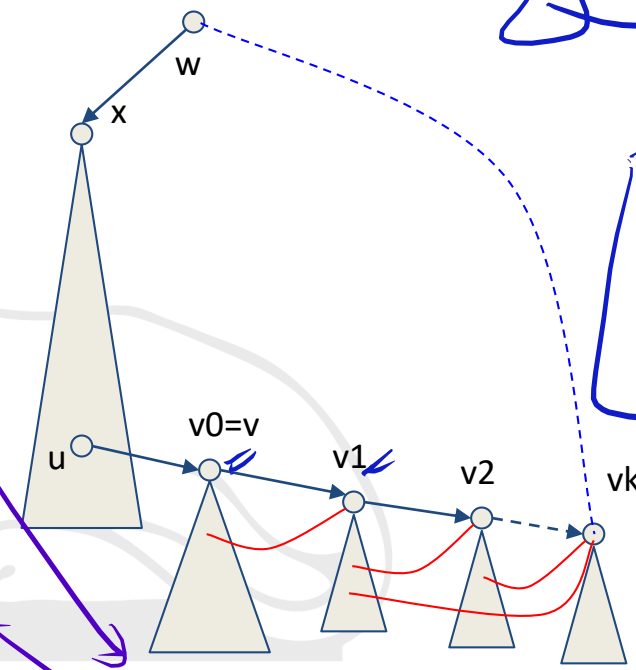
Number of edges

$\sum |T_i| (k-i)$

Fall of vertices

$\sum |T_i| (i)$

$\times (2i+1)$



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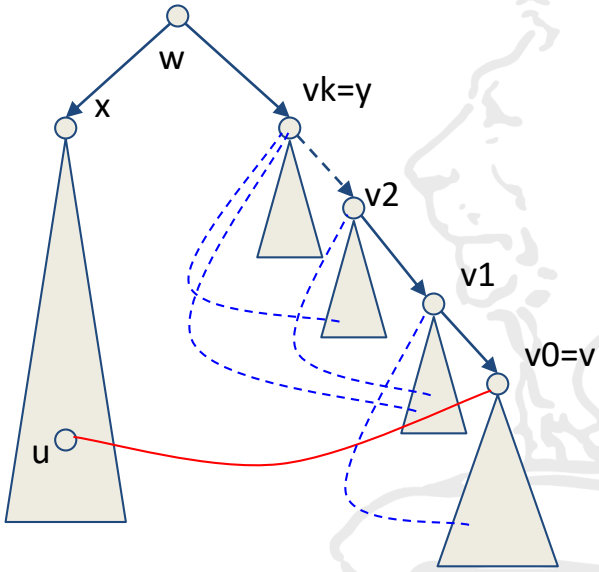

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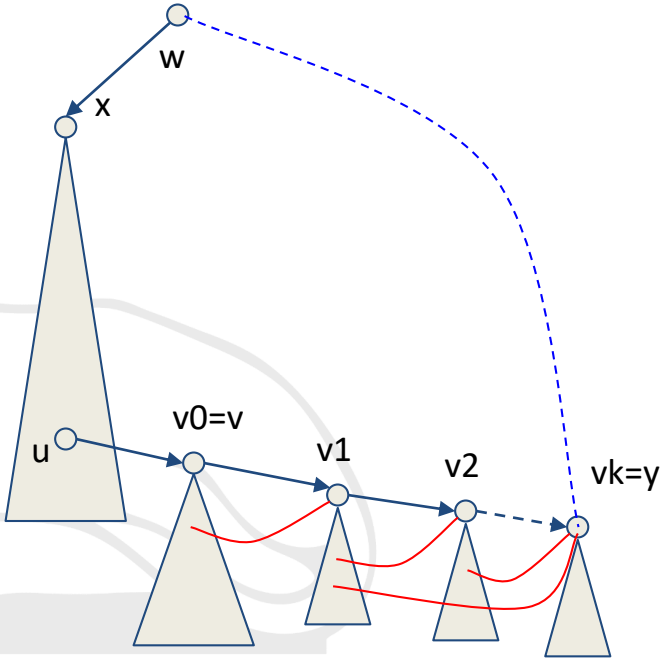
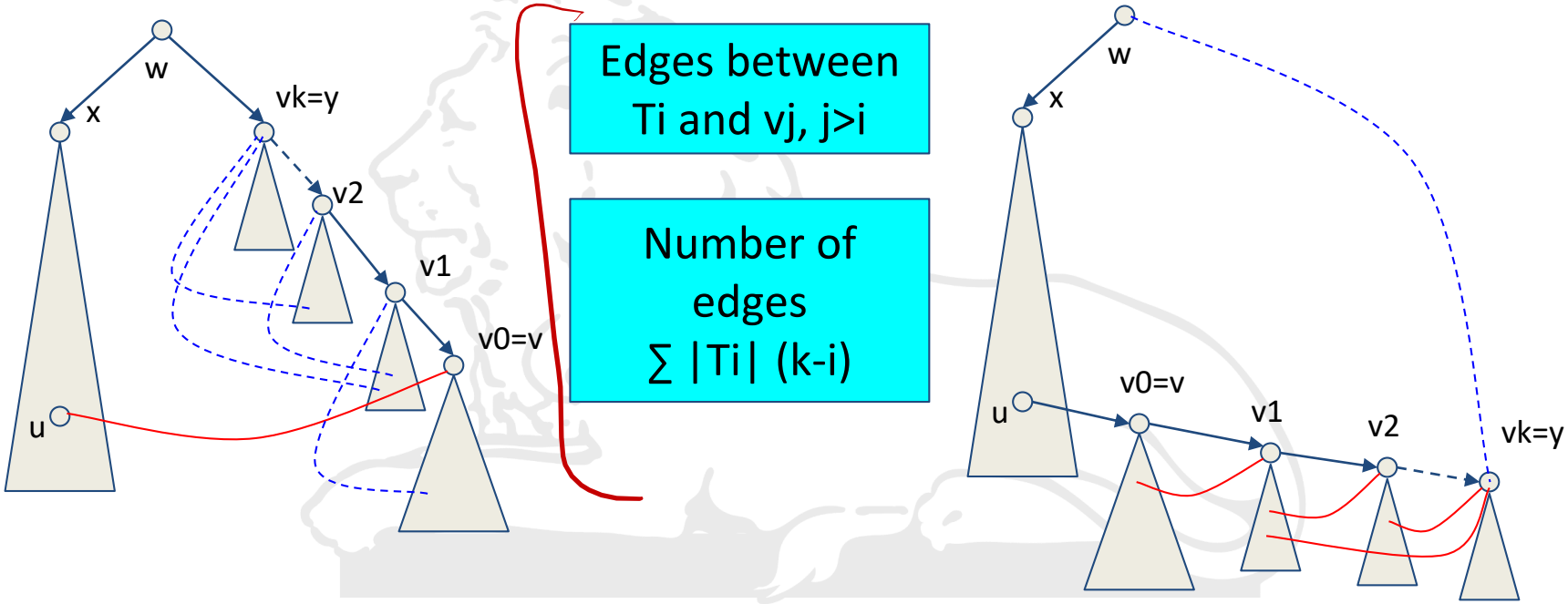


Processing **Cross Edge** Good because it leads to Fall

Back Edges converting to Cross Edge

Think?
How to efficiently collect such edges?



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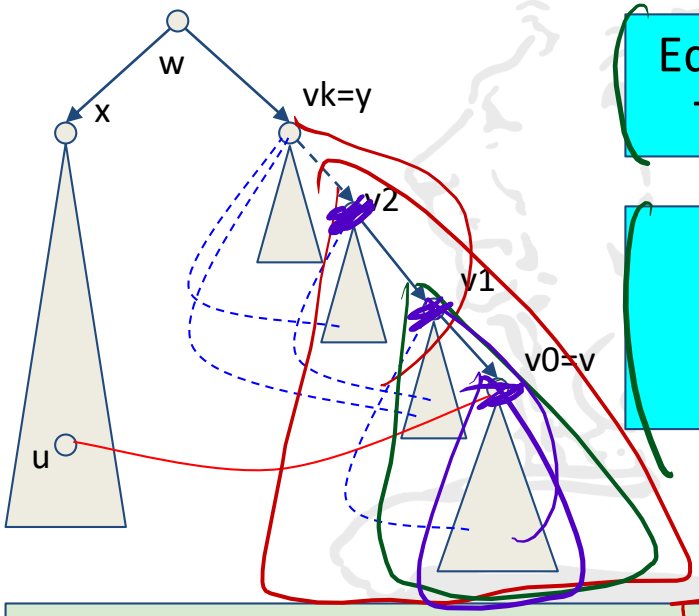

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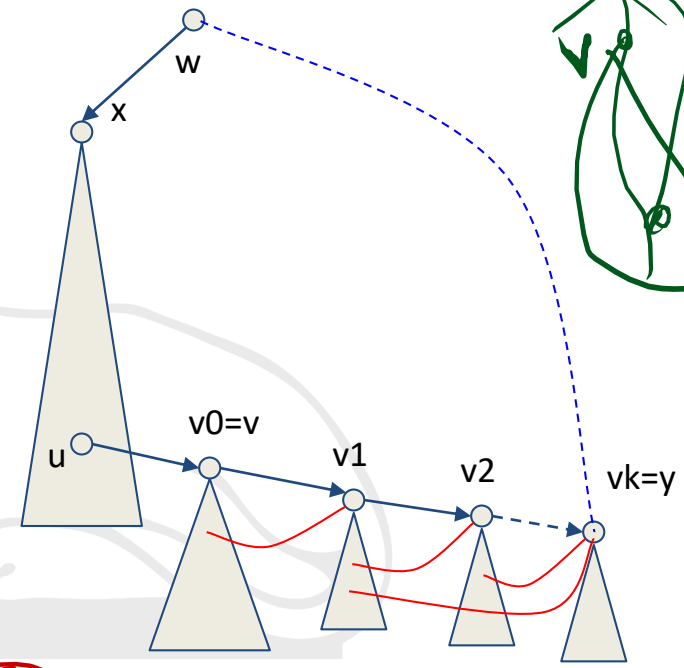

Each vertex v store back edges B[v]
- From the subtree T(v)
- To parent of v

Back Edges converting to Cross Edge

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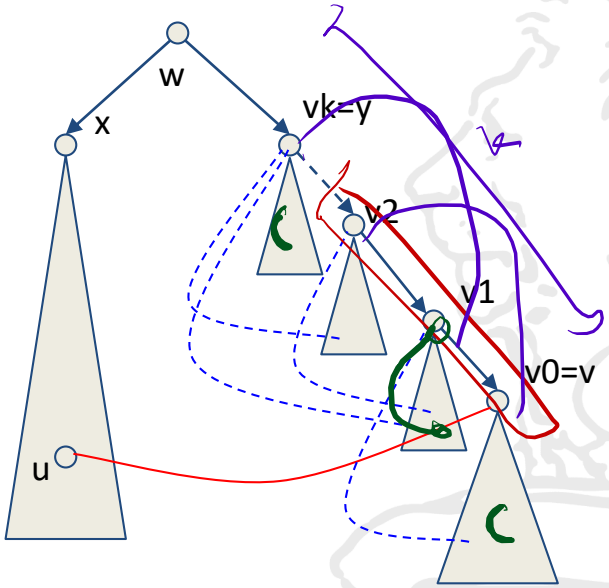

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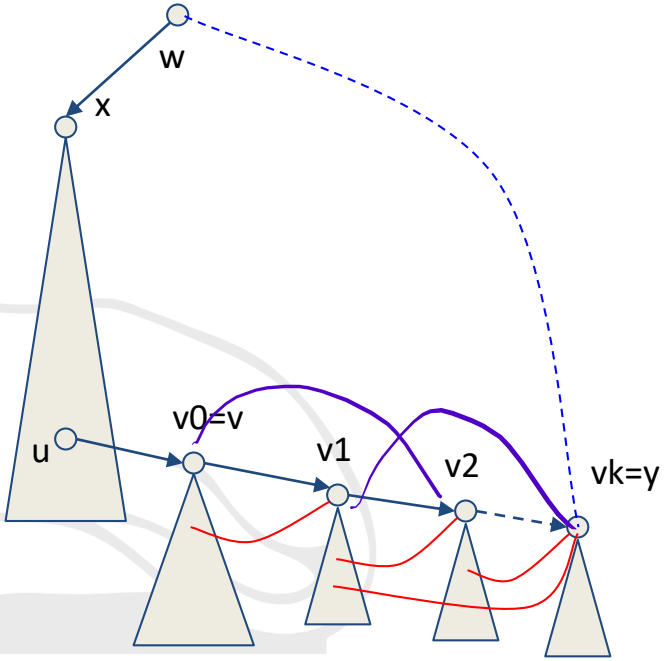

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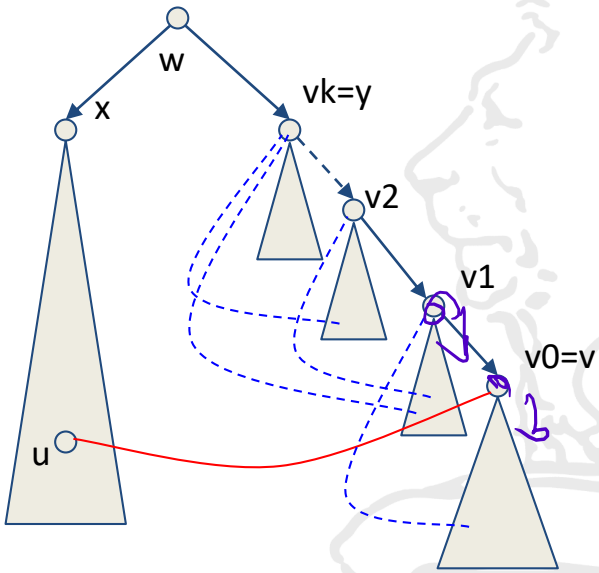
REQUIRED EDGES: B[x] for x in v->y {y}

Problem?

Back Edges converting to Cross Edge

Think?
How to efficiently collect such edges?



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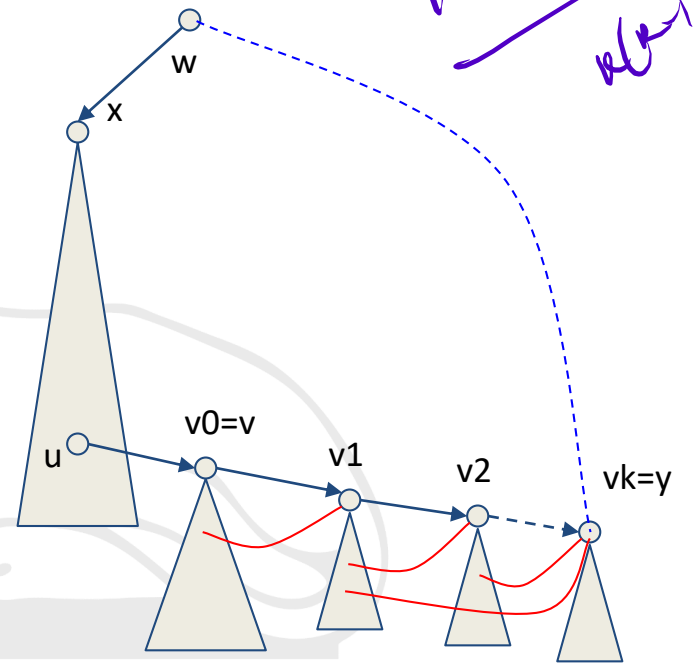

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k(k-1) back edges collected from v->y

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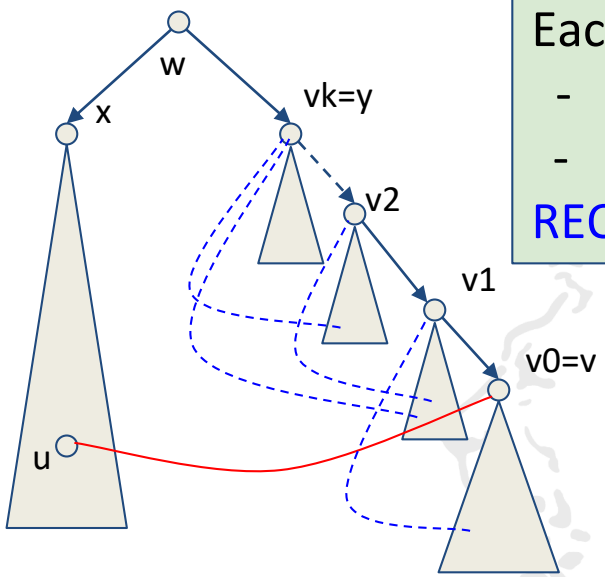

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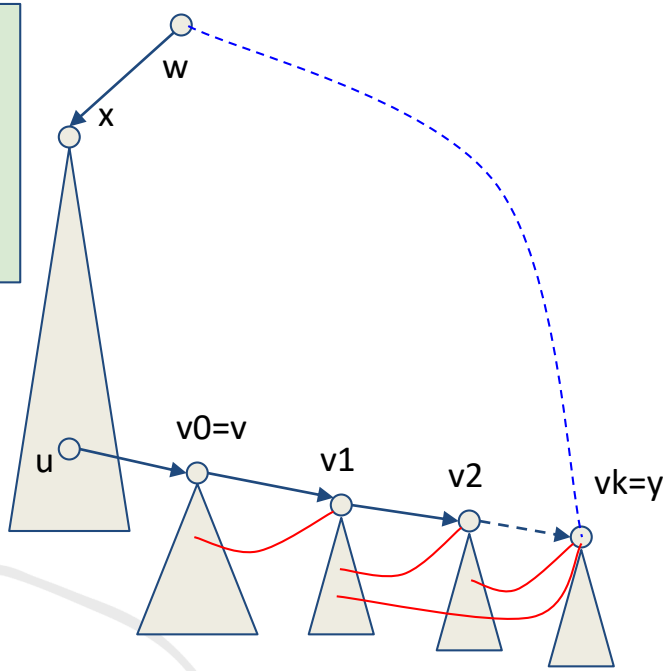


Target Analysis:

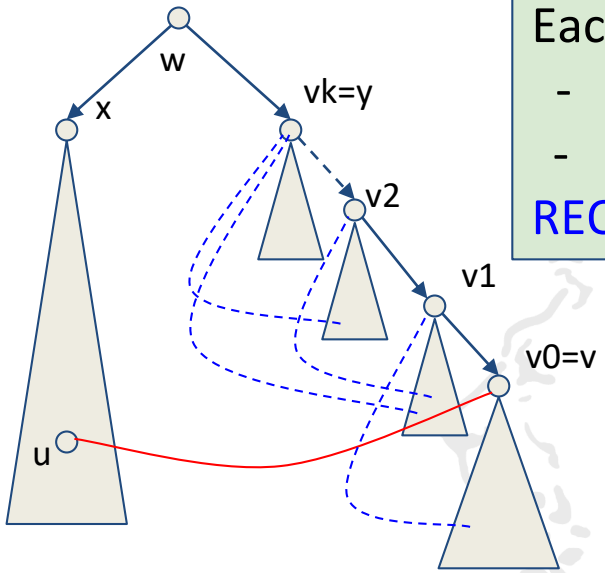
Charge O(1) whenever x falls

Total= n x O(n) = O(n2)

Processing **Cross Edge** Good because it leads to Fall

What about Back Edges?

Incremental Depth First Search Trees (Undirected)



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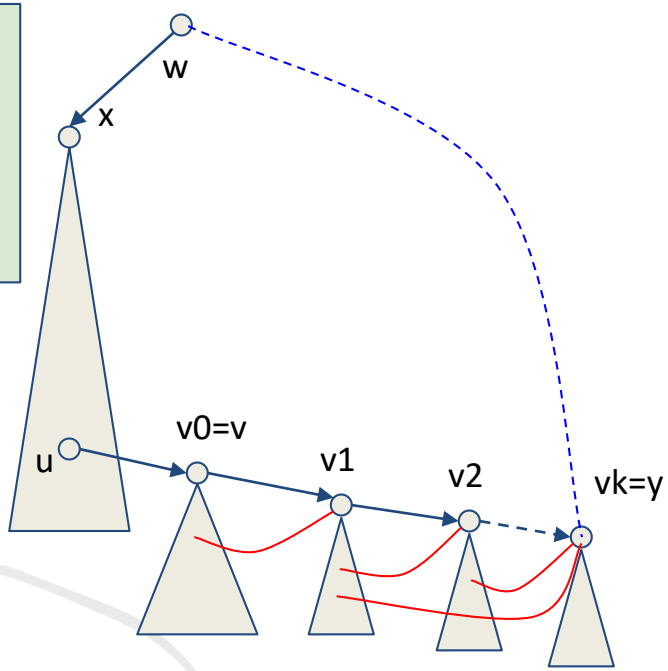

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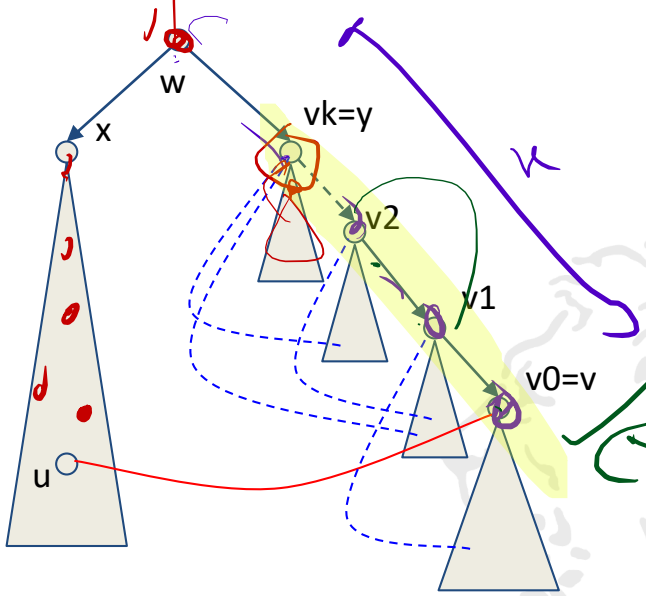
What about Back Edges?

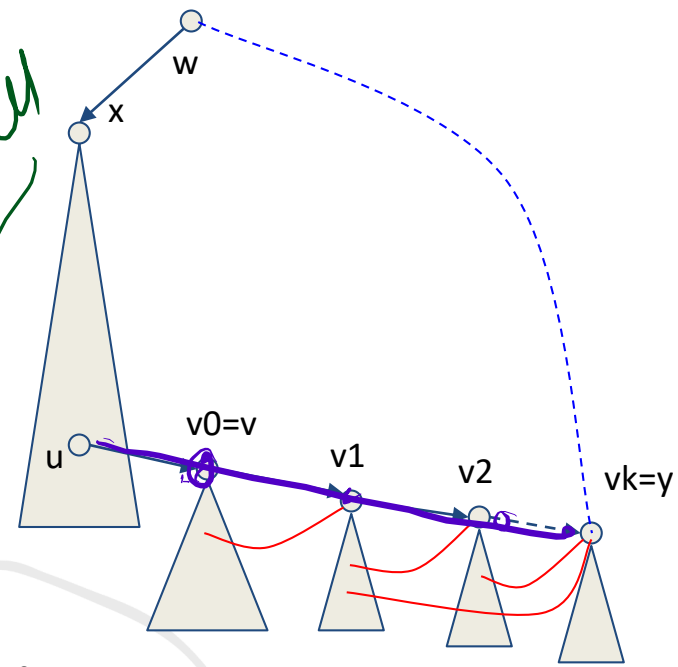
Minimum Fall? v->y path

1+2+...+k = k(k-1) levels

Incremental Depth First Search Trees (Undirected)



Update B[·] in some time to collect all



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- Collect Cross Edges => Fall

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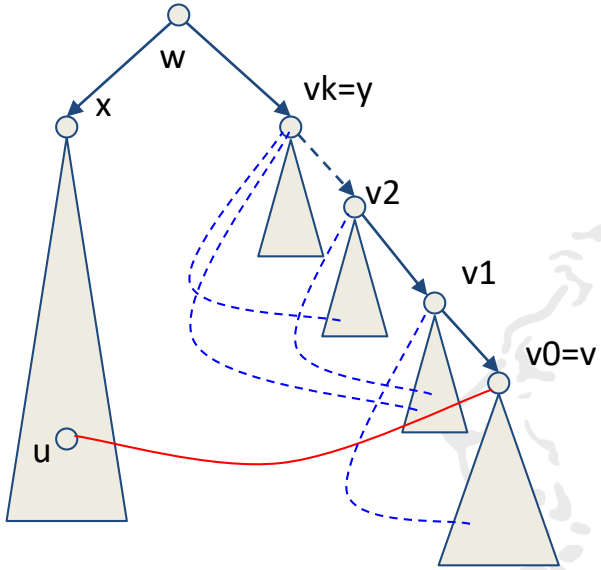

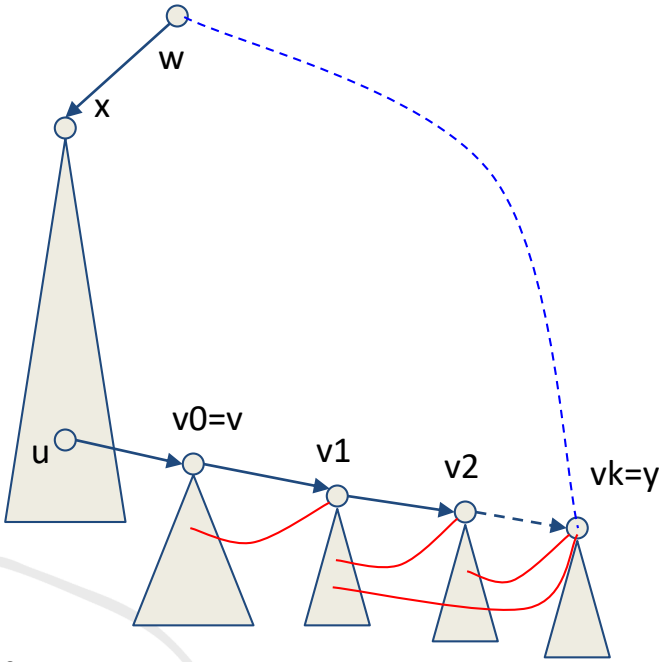

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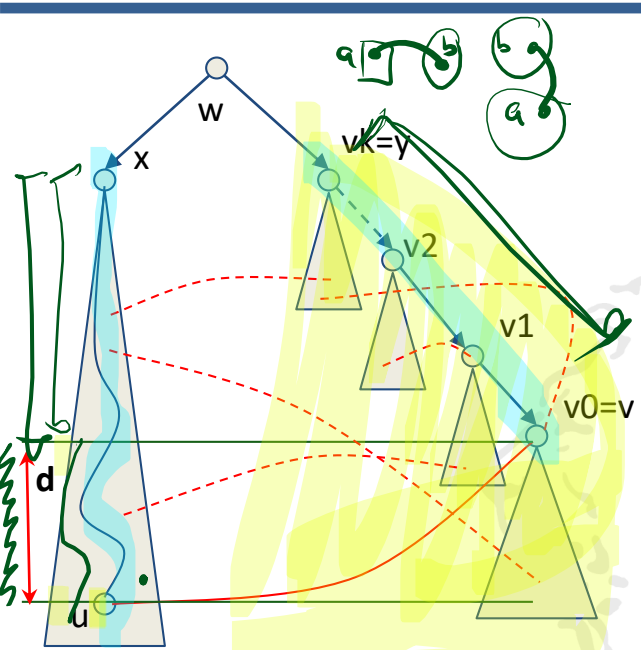
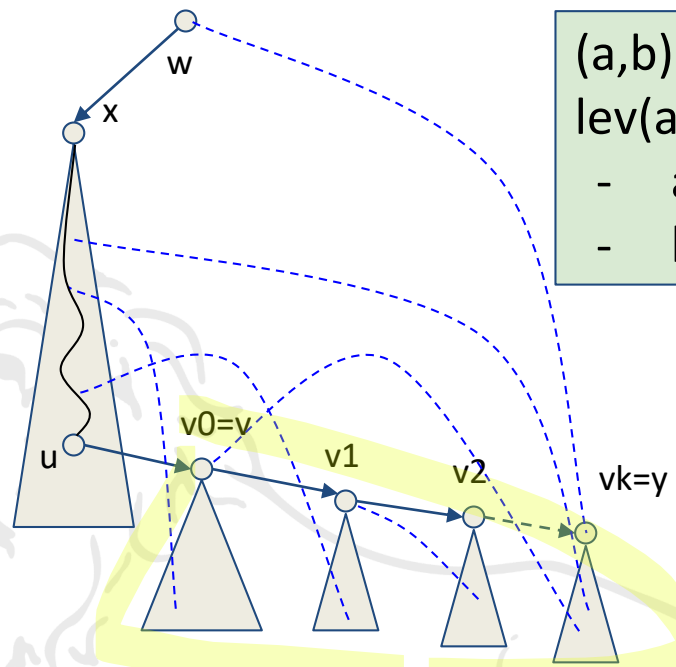
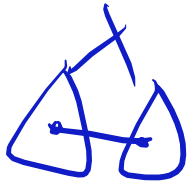
- Collect Back Edges
- Collect Cross Edges => Fall
- Collect Cross Edge => No fall?

Cross Edges Converting to Back Edges?





(a,b) converted?
lev(a)<lev(b)
- a is new ancestor of b
- b is new descendant of a



During above insertion

- Which vertices got new descendants? x->u and v->y
- Which vertices got new ancestors? T(v)
- Maximum conversions of cross edges? (|x->u|+|v->y|) * |T(v)|
- Minimum Fall in vertices?

(1+d)* |T(v)|

Minimum Fall >= **Maximum Conversions**?

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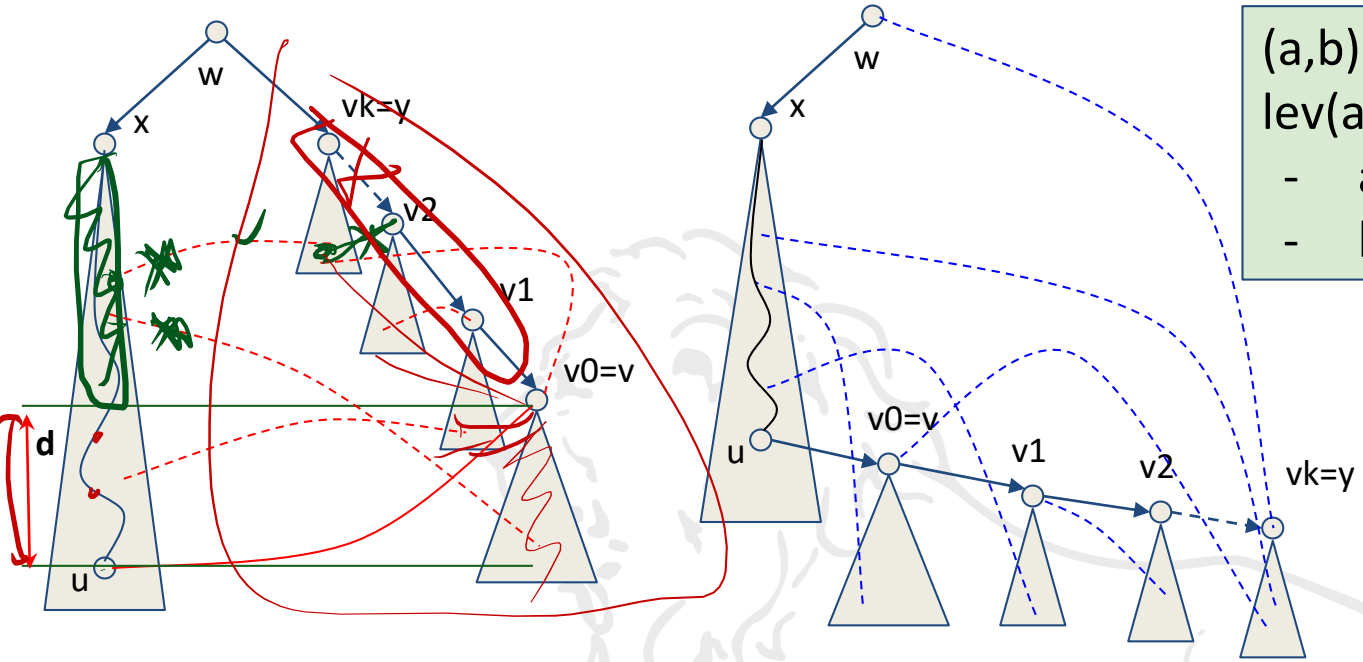

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IDEA
Process cross edges with highest end vertices first?

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$(1+d) * |T(v)|$

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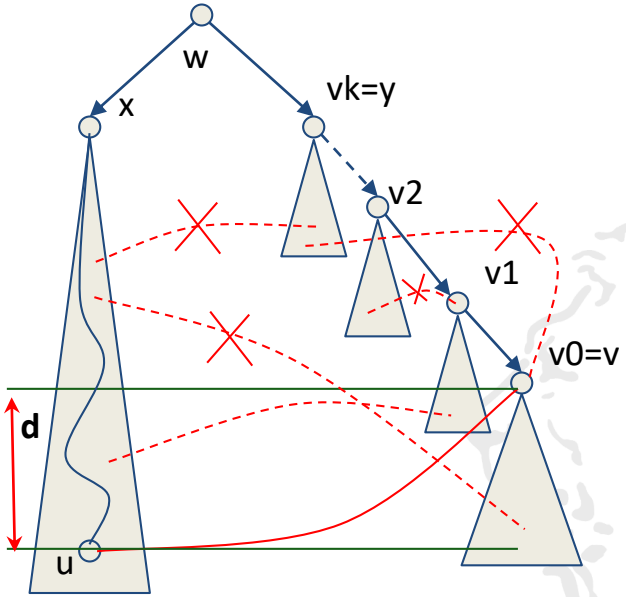

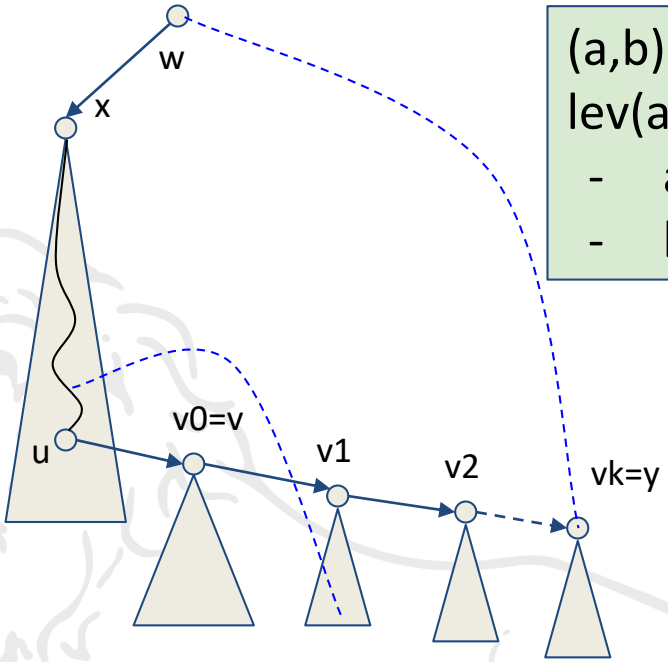

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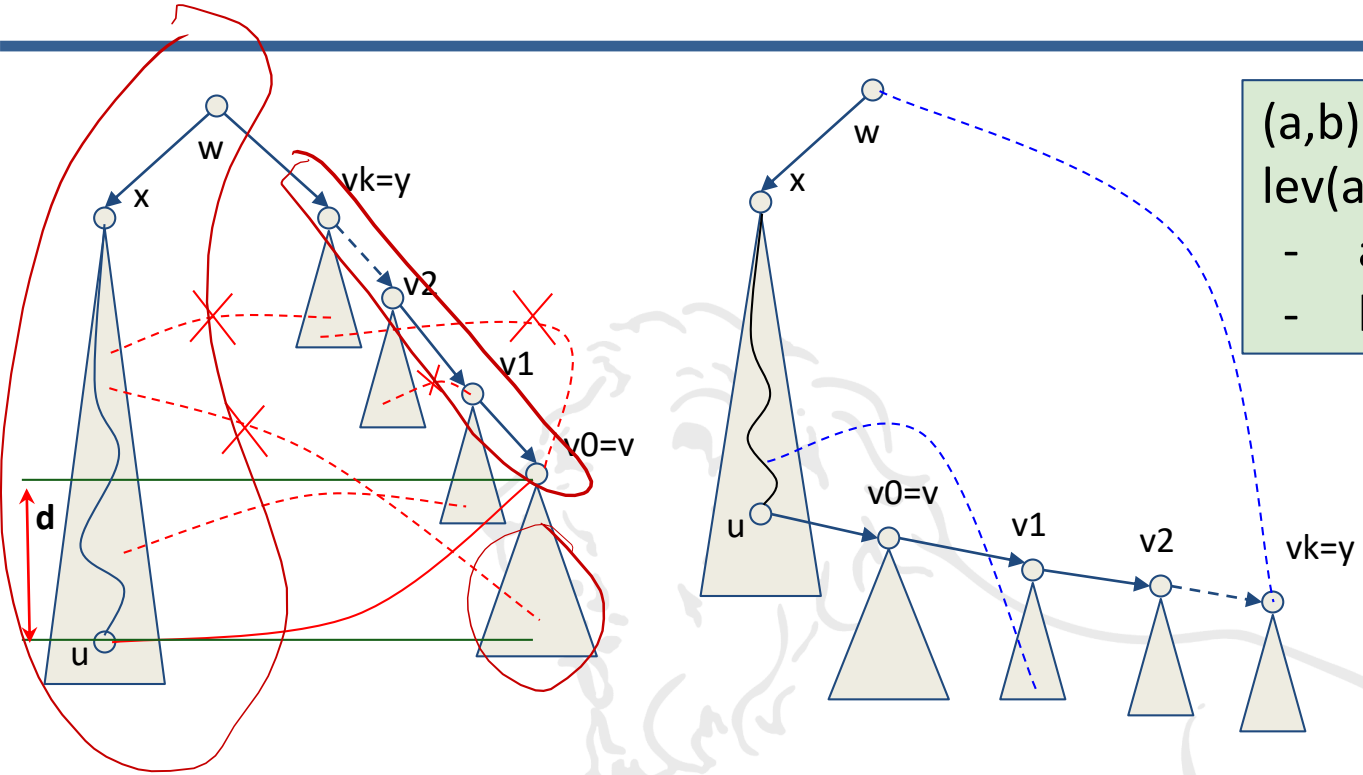

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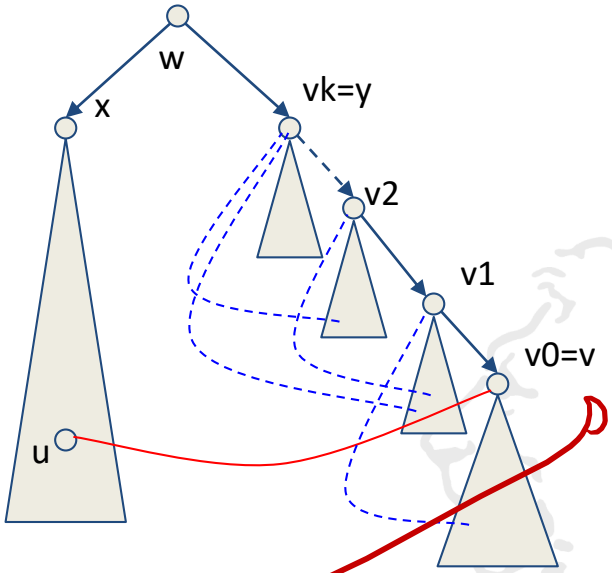
During above insertion

- Which vertices got new descendants? ~~x->u and v->y~~ u'->u and y?
- Which vertices got new ancestors? T(v)
- Maximum conversions of cross edges? d * |T(v)| + | T(v1)|?
- Minimum Fall in vertices? d * |T(v)| + |T(v1)|?

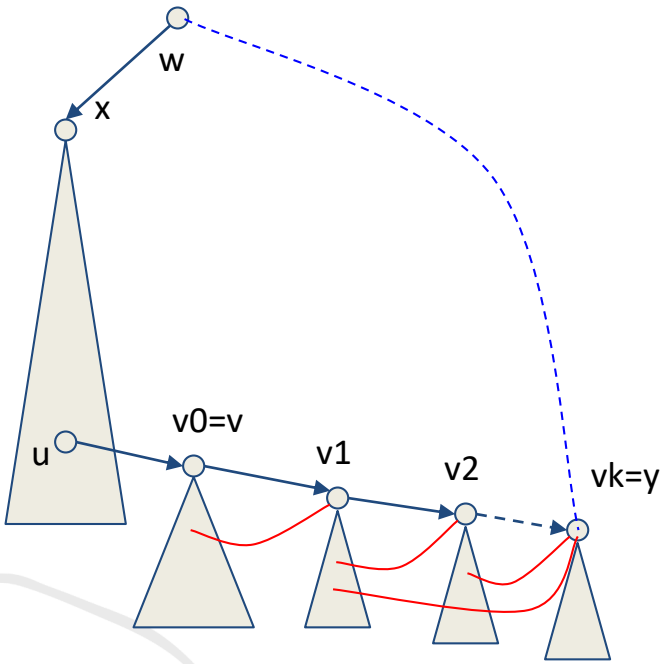
Minimum Fall >= **Maximum Conversions**?

Incremental Depth First Search Trees (Undirected)



IDEA
Process cross edges with highest end vertices first?

elu store in B[•]



Algorithm:

```
InsE = (u,v)
While(InsE not Empty)
    (u,v) = Pop highest from InsE
    Add (u,v) to G          // lev(u)<=lev(v)
    If( LCA(u,v)!={u,v})
        Reverse (v,LCA(u,v)) except LCA(u,v)
        Add (u,v) to DFS Tree
        Add cross edges of T(v) in InsE
        Update LCA structure
```

Analysis:

Charge O(1) whenever x falls

Total= n x O(n) = O(n2)

- Traverse and Reverse Path
- Collect Back Edges
- Collect Cross Edges => Fall
- Collect Cross Edge => No fall

Question

Think about **incremental maintenance of DFS** trees.

Directed acyclic graph **O(mn)**

Undirected **O(n2)**

Directed graph **O(m2)**

Paper Reference

Baswana Khan 14

Decremental DFS in DAGs
Ponder over how DFS of a DAG changes, on deleting an edge?